

Fixed width addition

Fixed-width addition: adding one bit at time, using the usual column-by-column and carry arithmetic, and dropping the carry from the leftmost column so the result is the same width as the summands. *Does this give the right value for the sum?*

$$\begin{array}{r} [2\ 6]_{10,2} \\ + [8\ 4]_{10,2} \\ \hline \end{array}$$

$$\begin{array}{r} (1\ 1\ 0\ 1\ 0\ 0)_{2,6} \\ + (0\ 0\ 0\ 1\ 0\ 1)_{2,6} \\ \hline \end{array}$$

$$\begin{array}{r} [1\ 1\ 0\ 1\ 0\ 0]_{s,6} \\ + [0\ 0\ 0\ 1\ 0\ 1]_{s,6} \\ \hline \end{array}$$

$$\begin{array}{r} [1\ 1\ 0\ 1\ 0\ 0]_{2c,6} \\ + [0\ 0\ 0\ 1\ 0\ 1]_{2c,6} \\ \hline \end{array}$$

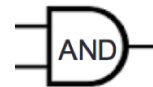
Can fixed-width addition be used to calculate $17 + (-17) = 0$? Hint: use 5 bits.

Circuits basics

In a **combinatorial circuit** (also known as a **logic circuit**), we have **logic gates** connected by **wires**. The inputs to the circuits are the values set on the input wires: possible values are 0 (low) or 1 (high). The values flow along the wires from left to right. A wire may be split into two or more wires, indicated with a filled-in circle (representing solder). Values stay the same along a wire. When one or more wires flow into a gate, the output value of that gate is computed from the input values based on the gate's definition table. Outputs of gates may become inputs to other gates.

Logic gates definitions

Inputs		Output
x	y	x AND y
1	1	1
1	0	0
0	1	0
0	0	0



Inputs		Output
x	y	x XOR y
1	1	0
1	0	1
0	1	1
0	0	0

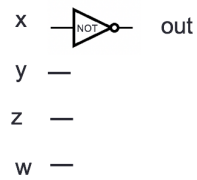


Input	Output
x	NOT x
1	0
0	1



Digital circuits basic examples

Example digital circuit:



Output when $x = 1, y = 0, z = 0, w = 1$ is _____

Output when $x = 1, y = 1, z = 1, w = 1$ is _____

Output when $x = 0, y = 0, z = 0, w = 1$ is _____

Draw a logic circuit with inputs x and y whose output is always 0. *Can you use exactly 1 gate?*